



Requirements Document: AI-Enabled Skills and Contribution Assessor

Project Overview

This project aims to create an AI-enabled system leveraging the GitHub Issue API and ChatGPT to summarize intern contributions and classify skills gained per issue in the Hack for LA organization. It generates resume-ready bullet points and labels skill sets from a predefined CSV taxonomy.

Functional Requirements

User Input and Interaction

- Users specify contributors one at a time through text entry.
- No authentication or verification is required.

Data Processing

- Fetch comments per issue for each specified user via GitHub Issue API.
- Extract context from issue overview and action items.
- Submit combined issue context and user comments to ChatGPT.

Output Generation

- **Bullet Points:**
 - Formal style with software development concepts.
 - Maximum of 50 words per bullet point.
 - Explicitly state task status (completed or ongoing).
- Publicly posted as a comment to the original GitHub issue.
- **Skill Classification:**
 - Based on provided CSV file (`HfLA_skills.csv`).
 - Output multiple skill labels relevant to each user's contributions.

Non-Functional Requirements

Performance

- The entire operation per issue per user must complete within 1 minute.

Reliability and Error Handling

- Graceful error handling with public posting of error code and message to the corresponding GitHub issue upon API failure (GitHub or ChatGPT).

Security

- No specific encryption or anonymization required.
- No anticipated privacy concerns.

Deployment and Environment

- Currently no existing deployment infrastructure.
- Designed for scheduled daily execution across all open issues within Hack for LA organization, incrementing through issues sequentially.
- Codebase will reside in GitHub repo: [hackforla/ai-skills-assessor](https://github.com/hackforla/ai-skills-assessor).
- Data persistence will use GitHub as the database (folder: `/Data`).

Scalability

- Designed for low concurrent load given daily, sequential batch processing.

Usability

- No graphical user interface; automated execution through daily time-triggered scripts.

Data and API Requirements

APIs

- **GitHub Issue API:**
 - Used to retrieve user comments and issue context.
 - Respect GitHub API rate limits (to be researched and managed in implementation).
- **ChatGPT API:**
 - Premium ChatGPT API key to be acquired.
 - Usage within API limits to be researched and respected.

Data

- Skill taxonomy sourced from provided CSV file (`HfLA_skills.csv`), structured as comma-separated skill labels without newlines.
- All relevant issues and comments pre-existing within GitHub.

Feasibility Considerations

- **Technical Feasibility:**

- Ensure API response times and limits from GitHub and OpenAI are manageable.
- Initial prototyping to validate response times and API usage efficiency.
- **Financial Feasibility:**
 - Consider costs associated with premium ChatGPT API usage.

Development Stack

- JavaScript development building on existing codebase.
- Leverage existing true-github-contributors package, modifying it for AI-enabled summarization and skill classification tasks.

This requirements document serves as the foundation for subsequent project development, providing clear guidelines for implementation, validation, and future enhancements.